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3.1 Translations

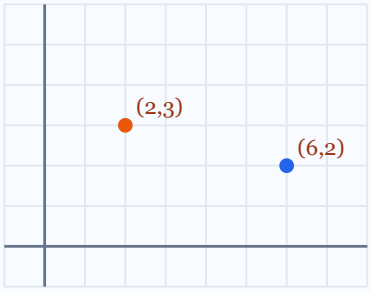
A translation slides every point of a figure the same distance in the same direction, and that single shared shift is why translations are rigid motions — distance, angle measure, and orientation all survive intact. This section goes past applying a rule to one point: it asks you to prove algebraically why distance is preserved, translate an entire triangle vertex by vertex, work backward from an image to a preimage, and combine two translations into one equivalent vector.

Examples

Example 1: Translate a point using vector notation

Translate $(2, 3)$ by $\langle 4, -1 \rangle$. Find the image point.

Solution



Every point shifts by the same vector $\langle 4,-1 \rangle$.

A translation vector $\langle a, b \rangle$ shifts every point by adding a to the x-coordinate and b to the y-coordinate.

$(2 + 4,\ 3 + (-1))$.

Answer: $(6, 2)$.

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Example 2: Apply a translation rule

The rule $(x, y) \to (x + 3, y - 5)$ translates a figure. Find the image of $(1, 8)$.

Solution

Substitute the point's coordinates directly into the rule: $x = 1 \Rightarrow x + 3 = 4$; $y = 8 \Rightarrow y - 5 = 3$.

Answer: $(4, 3)$.

Example 3: Find the vector, then apply it elsewhere

$(1, 2)$ maps to $(5, 6)$ under a translation. Find the vector, then use it to find the image of $(0, 0)$.

Solution

Subtract coordinates to see how far the point moved: vector $= \langle 5 - 1,\ 6 - 2 \rangle = \langle 4, 4 \rangle$.

Apply that same vector to $(0, 0)$: $(0 + 4,\ 0 + 4)$.

Answer: vector $\langle 4, 4 \rangle$; the image of $(0, 0)$ is $(4, 4)$.

Example 4: Why translations preserve distance

Explain, using coordinates, why a translation never changes the distance between two points.

Solution

Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$, translated by $\langle a, b \rangle$ to $P_1' = (x_1 + a,\ y_1 + b)$ and $P_2' = (x_2 + a,\ y_2 + b)$.

The horizontal difference becomes $(x_2 + a) - (x_1 + a) = x_2 - x_1$ — the a's cancel; the vertical difference cancels the same way.

Since both coordinate differences are unchanged, the distance formula gives the exact same result before and after.

Answer: The shared shift cancels out when you subtract to find the difference, so distance is always preserved.

Example 5: Orientation is preserved

Does a translation reverse a figure's orientation (the clockwise vs. counterclockwise order of its vertices)? Explain.

Solution

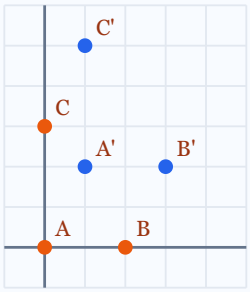
Orientation depends only on the rotational order of the vertices, not on where the figure sits — sliding a shape without turning or flipping it cannot swap clockwise for counterclockwise.

Answer: No — translations preserve orientation (unlike reflections, which reverse it).

Example 6: Translate a whole triangle

$\triangle ABC$ has $A(0,0)$, $B(2,0)$, $C(0,3)$. Translate by $\langle 1,2\rangle$. Find all three image vertices.

Solution



Every vertex shifts by the same vector ⟨1,2⟩.

Add $\langle 1,2\rangle$ to every vertex: $A' = (0+1, 0+2)$; $B' = (2+1, 0+2)$; $C' = (0+1, 3+2)$.

Answer: $A'(1,2)$, $B'(3,2)$, $C'(1,5)$ — same size and shape as $\triangle ABC$, just shifted.

Example 7: Decompose a slide into left/right and up/down

Describe the translation $(x,y) \to (x-4, y+6)$ using direction words.

Solution

The x-coordinate decreases by 4: shift **left** 4. The y-coordinate increases by 6: shift **up** 6.

Answer: Left 4 units and up 6 units — equivalent to vector $\langle -4, 6\rangle$.

Example 8: Find the preimage

Under the rule $(x, y) \to (x + 2,\ y - 3)$, a point P maps to $(2, 4)$. Find P.

Solution

Reverse the rule by subtracting instead of adding: $x = 2 - 2 = 0$; $y = 4 - (-3) = 7$.

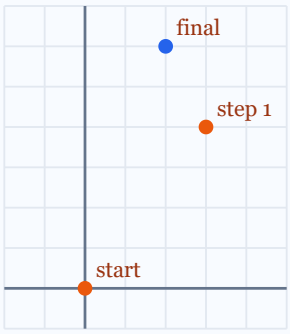
Check: $(0 + 2,\ 7 - 3) = (2, 4)$. ✓

Answer: $P = (1, 7)$.

Example 9: Combine two translations

A robot starts at $(0, 0)$. It moves by vector $\langle 3, 4\rangle$, then by vector $\langle -1, 2\rangle$. Find its final position, and the single vector that would take it there directly.

Solution



Two moves in a row equal one combined vector.

Apply the first move: $(0, 0) + \langle 3, 4\rangle = (3, 4)$.

Apply the second move: $(3, 4) + \langle -1, 2\rangle = (2, 6)$.

A single equivalent vector is the sum of the two: $\langle 3, 4\rangle + \langle -1, 2\rangle = \langle 2, 6\rangle$, matching the direct path from $(0, 0)$ to $(2, 6)$.

Answer: Final position $(2, 6)$; combined vector $\langle 2, 6\rangle$.

Example 10: Midpoint travels along with the segment

$\overline{AB}$ has midpoint M. If the segment is translated by $\langle a, b\rangle$, show that the image of M is the midpoint of the image segment $\overline{A'B'}$.

Solution

Let $A = (x_1, y_1)$, $B = (x_2, y_2)$, so $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$.

After translating, $A' = (x_1 + a,\ y_1 + b)$ and $B' = (x_2 + a,\ y_2 + b)$, so the midpoint of $\overline{A'B'}$ is $(\frac{x_1 + x_2}{2} + a,\ \frac{y_1 + y_2}{2} + b)$.

That is exactly M shifted by $\langle a, b\rangle$ — the midpoint moves along with the segment.

Answer: The image of the midpoint equals the midpoint of the image; translations preserve midpoints.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Translate

$(3, 1)$ by $\langle 2, 5 \rangle$?

Answer: **Verification:**

Problem 2: Rule

$(x, y) \to (x - 3, y + 4)$; image of $(4, 0)$?

Answer: **Verification:**

Problem 3: Vector

$(0, 0)$ to $(3, -2)$. Vector?

Answer: **Verification:**

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Problem 4: Rigid?

Is a translation a rigid motion?

Answer: **Verification:**

Problem 5: Orientation

Does translation reverse orientation?

Answer: **Verification:**

Problem 6: Image

$(-1, 5)$ by $\langle 4, -3 \rangle$?

Answer: **Verification:**

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Problem 7: Direction

$(x, y) \to (x + 7, y)$ moves how?

Answer: **Verification:**

Problem 8: Direction

$(x, y) \to (x, y - 2)$ moves how?

Answer: **Verification:**

Problem 9: Preimage

Rule $(x + 1, y - 3)$; image $(2, 4)$. Preimage?

Answer: **Verification:**

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Problem 10: Triangle

$A(1, 1)$ by $\langle -2, 3 \rangle$?

Answer: **Verification:**

Problem 11: Vector

$(4, 7)$ to $(4, 1)$. Vector?

Answer: **Verification:**

Problem 12: Congruence

Is a figure congruent to its translate?

Answer: **Verification:**

Problem 13: Rule

Translate left 5, up 2: rule?

Answer: **Verification:**

Problem 14: Challenge — translate a triangle

Triangle $D(-1, 2)$, $E(3, 2)$, $F(1, -2)$ is translated by $\langle -2, 5 \rangle$. Find all three image vertices.

Answer: **Verification:**

Problem 15: Challenge — midpoint two ways

Segment endpoints $(2, -3)$ and $(6, 1)$ have midpoint $(4, -1)$. Translate by $\langle -5, 2 \rangle$. Find the new midpoint (a) by translating the midpoint directly and (b) by translating both endpoints and averaging. Confirm they match.

Answer: **Verification:**

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✓ Answer Key

1. $(5, 6)$
2. $(1, 4)$
3. $\langle 3, -2 \rangle$
4. Yes
5. No
6. $(3, 2)$
7. Right 7
8. Down 2
9. $(1, 7)$
10. $(-1, 4)$
11. $\langle 0, -6 \rangle$
12. Yes
13. $(x, y) \to (x - 5, y + 2)$
14. $D'(-3, 7), E'(1, 7), F'(-1, 3)$
15. Both methods give $(-1, 1)$

Great work — keep practising!